Flange Nozzle Selection

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1. Purpose

The document describes the criteria and the procedure to define the dimensions of centrifugal pump flanges.

The spreadsheet on Mod.463 calculates the suction and discharge flange dimensions, based on several acceptance criteria and double checks any Customer request, generally declared in datasheet.

2. Scope.

This procedure is applicable to all the centrifugal pumps.

3. Abbreviations

N	*[rpm]*	*pump rotor speed*
Qbep	*[m3/h or GPM]*	*pump flow rate at best efficiency point and maximum diameter*
Hst.bep	*[m or ft]*	*pump head per stage at best efficiency point and maximum diameter*
D2	*[mm or in]*	*maximum external diameter of the impeller*
Ld	*[mm or in]*	*length of the diffusive duct*
Qbleedoff	*[m3/h or GPM]*	*bleed off flow rate*
D1	*[mm or in]*	*impeller eye diameter*
Dh	*[mm or in]*	*shaft impeller diameter*
Din	*[mm or in]*	*suction flange diameter*
Dout	*[mm or in]*	*discharge flange diameter*
Vin	*[m/s or ft/s]*	*velocity on the suction flange*
Vout	*[m/s or ft/s]*	*velocity on the discharge flange*
Loss/Hst	*[%]*	*diffusive duct loss as percentage of Head per stage of the pump*
KE/Hst	*[%]*	*kinetic energy as percentage of Head per stage of the pump*
Nq or Ns	*[/]*	*specific speed of the pump*

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4. General procedure: description.

The procedure furnishes an estimation of the diameter of the suction and discharge flanges of a pump, based on the following criteria:

- Standard Velocity ranges [1]
- Kinetic energy [1]
- Diffusion Losses [2]

4.1. Standard Velocity ranges

Typical velocity limits on the flanges are:

- Vin < 4.6 m/s [or 15 ft/s]
- Vout <7.6 m/s [or 25 ft/s]

For VS type pumps the limits are lowered down to:

- Vin from 3.3 m/s [or 11 ft/s] to 4.6m/s [or 15 ft/s]
- Vout from 3.3 m/s [or 11 ft/s] to 4.6 m/s [or 15ft/s]

4.2. Kinetic energy

Kinetic energy is the amount of energy related to the velocity of the flow at the inlet and at the outlet of the pump. it's calculated using the formula:

$$KE = Vx^2/2g$$

where:

- Vx is the velocity at the inlet or at the discharge of the pump flanges.
- G is the acceleration of gravity.

Typical limit for the kinetic energy on the pump flanges is expressed below:

$$KE/Hst < 3\%$$

4.3. Diffusion Losses

The calculation of the losses in the diffusive duct are strictly related to the ratio of the area at the inlet of the diffusive duct (typically the throat area) and the area in the outlet of the duct (that is based on the diameter of the discharge flange).

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The formula used for the calculation are in the Table 3.8 (2) of [2]:

Table 3.8 (2) Hydraulic losses in diffuser or volute					Eq.
Hydraulic efficiency		$\eta_h = \frac{Y}{Y_{sch}} = \frac{H}{H_{th}} = \frac{H}{H + Z_{La} + Z_{Le} + Z_{EA}} = \frac{\psi}{\psi + \zeta_{La} + \zeta_{Le} + \zeta_{EA}}$			3.8.1
Diffuser	Velocity vector at impeller outlet	$c_{2u} = \frac{gH}{\eta_h u_2} + d^*_{1m} c_{1u}$	$c_{2m} = \frac{Q_{La}}{\pi d_2 b_2}$	$c_2 = \sqrt{c_{2u}^2 + c_{2m}^2}$	3.8.15
Diffuser	Flow velocity in throat	$c_{3q} = \frac{Q + Q_E}{z_{Le} a_3 b_3}$			3.8.16
Diffuser	Diffuser c_p from Fig. 1.18 or 1.19	$A_R = \frac{a_4 b_4}{a_3 b_3}$	$\frac{L_{3-4}}{R_1} = L_{3-4}\sqrt{\frac{\pi}{a_3 b_3}}$	$c_p = f\left(A_R, \frac{L_{3-4}}{R_1}\right)$	3.8.17
Diffuser	Friction losses in inlet region	$\zeta_{2-3} = \frac{2g Z_{2-3}}{u_2^2} = (c_f + 0.0015)\left(a_3^* + b_3^*\right)\frac{\pi^3 \left(\varphi_2 b_2^*\right)^2}{8\left(z_{Le} a_3^* b_3^*\right)^3}\left(1 + \frac{c_2}{c_{3q}}\right)^3$			3.8.18
Diffuser	Diffuser loss including overflow channel	$\zeta_{Le} = \frac{2g Z_{Le}}{u_2^2} = \zeta_{2-3} + \left(\frac{c_{3q}}{u_2}\right)^2\left\{0.3\left(\frac{c_2}{c_{3q}} - 1\right)^2 + 1 - c_p - \frac{1 - \zeta_{ov}}{A_R^2}\right\}$			3.8.19
Diffuser	Overflow channel (alone)	$\Delta\zeta_R = \zeta_{ov}\frac{c_4^2}{u_2^2}$	$\zeta_{ov} = 0.2$ to 1.5		3.8.20
Volute	Friction loss ΔA= wetted surface	$\zeta_{Sp,R} = \frac{2gZ_{Sp}}{u_2^2} = \frac{2P_d}{\rho Q u_2^2} = \frac{1}{Q u_2^2}\Sigma\left(c_f + 0.0015\right)c^3\Delta A$			3.8.21
Volute	Diffuser/discharge nozzle (c_x = diffuser inlet velocity)	$\zeta_{Sp,D} = \frac{c_x^2}{u_2^2}\left(1 - c_p - \frac{1}{A_R^2}\right)$			3.8.22
Volute	Losses of volute including discharge nozzle	$\zeta_{Sp} = \zeta_{Sp,R} + \zeta_{Sp,D} + \zeta_{LS} + (\zeta_{Le} - \zeta_{Le,min})$ ζ_{LS} from Eq. (T3.8.24)			3.8.22a
Vaneless diffuser	Friction in vaneless diffuser with constant width	$\zeta_{LR} = \frac{2 c_f r_2}{b_3 \sin\alpha_3 \cos^2\alpha_3}\left(\frac{c_{2u}}{u_2}\right)^2\left(1 - \frac{r_2}{r_4}\right)$			3.8.23
Vaneless diffuser	Shock losses	$\zeta_{LS} = \varphi_{2,La}^2\left(\tau_2 - \frac{b_2}{b_3}\right)^2$			3.8.24
Deceleration and shock losses in vaned diffusers and volutes		$\zeta_{Le} - \zeta_{Le,min} = a + b\frac{c_{3q}}{c_2} + c\left(\frac{c_{3q}}{c_2}\right)^2$	volutes $c = 0.63$ $x_o = 0.87$ $a = c\,x_o^2$ $b = -2\,x_o\,c$	diffusers $c = 0.65$ $x_o = 0.93$ $a = c\,x_o^2$ $b = -2\,x_o\,c$	3.8.25
Add losses of vaneless space between impeller and diffuser or volute.					

For the calculation of the throat area data from typical velocity coefficient proposed by Stepanoff [3] are used. As shown in Figure 1. The chart relates the velocity coefficient with the specific speed of the pump:

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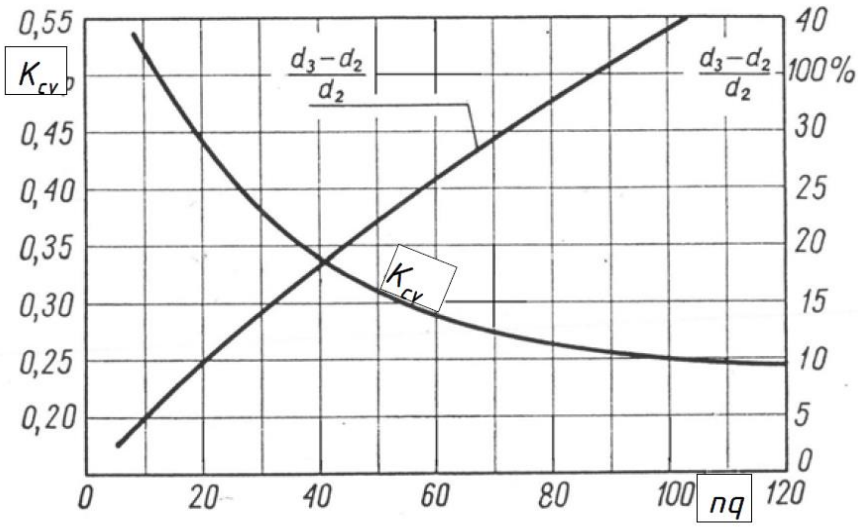

Figure 1. estimation of velocity coefficient kcv of the throat area velocity suggested by [3]

When kcv is defined, the throat velocity is calculated using the formula from [3]

Vth =kcv*(2*g*Hst)^0.5

The throat area will be calculated using the formula

Ath=Qbep/vth

The tool extracts the value of pressure recovery (cp) of the pump diffusive duct, using the chart proposed by [2], shown in Figure 2. The chart provides the value of cp as a function of area ratio (Aout/Athroat) and relative length of the diffusive duct (Ld/(Athroat/3.14)^0.5).

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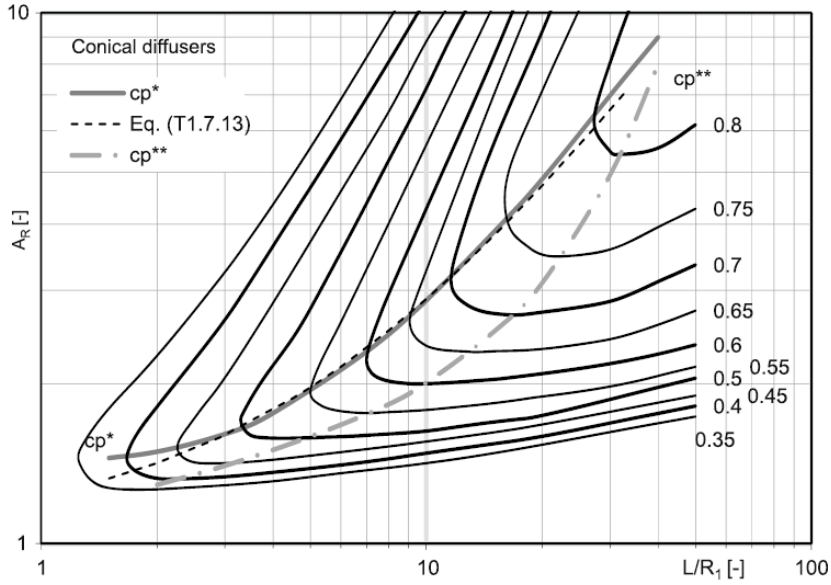

Figure 2. pressure recovery (Cp) as function of L/R1 and Ar

5. Description of the calculation tool.

The tool in Mod.463 estimates the velocities and the losses on the pump flanges and suggest the selection to achieve each of the criteria described in chapter 4.

5.1. Input data.

The input data of the tool are

- the performance of the pump at best efficiency point condition (n, Q, Hst)
- the VS pump type selection
- the main dimension of the impeller (D2, D1, Dh)
- approximative length of the diffusive duct Ld
- The selection of the suction and discharge flange (Din and Dout)

The mask of the input data is shown in Figure 3

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INPUT DATA		
unit system	SI	
Pump speed	1800	RPM
Qbep	4000	m3/h
Hpeb/stage	400	m
number of suctions	1	
D2	700.0	mm
Ld - Length of diffusive duct (approx.)	1900.0	mm
Qbleed off	0.0	m3/h
VS pump?	y	
D1	350.0	mm
Dh	150.0	mm

selected Diameter		
Din	600	mm
Dout	600	mm

Figure 3. Input data.

5.2. Output data

The output data of the tool furnishes the suggested dimension of suction and discharge flanges that fulfils the criteria described in chapt.4.

For each proposed diameter the following information are provided:

- Velocity
- Losses of diffusive duct as percentage of Head per stage
- Kinetic energy contributes as percentage of head per stage.

The output data are shown in Figure 4:

OUTPUT DATA

suggested values for outlet flange					
	for max eff	minimum	maximum	selected	
Diameter	416.7	600	700	600	mm
velocity	8.1	3.9	2.9	3.9	m/s
Loss/Hst	3.6%	5.3%	6.3%	5.3%	
KE/Hst	0.8%	0.2%	0.1%	0.2%	

Nq	Ns
21.2	1094.60

suggested values for inlet flange					
		minimum	maximum	selected	
Diameter		600	600	600	mm
velocity		3.9	3.9	3.9	m/s
KE/Hst		0%	0%	0%	

Figure 4. output data tables.

Based on the above suggests the user select the propre diameters for both the pump flanges.

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The tool will provide the value of velocity, losses and kinetic energy for the selection and shows in a graphical way if the selection is in or out of the criteria previously described.

An example of the graphical mode is shown in Figure 5. The red point refers to the selected solution, while the horizontal line refers to the ranges proposed by each method. The dashed lines indicated the velocity trend for the velocity by changing the dimension of the flange.

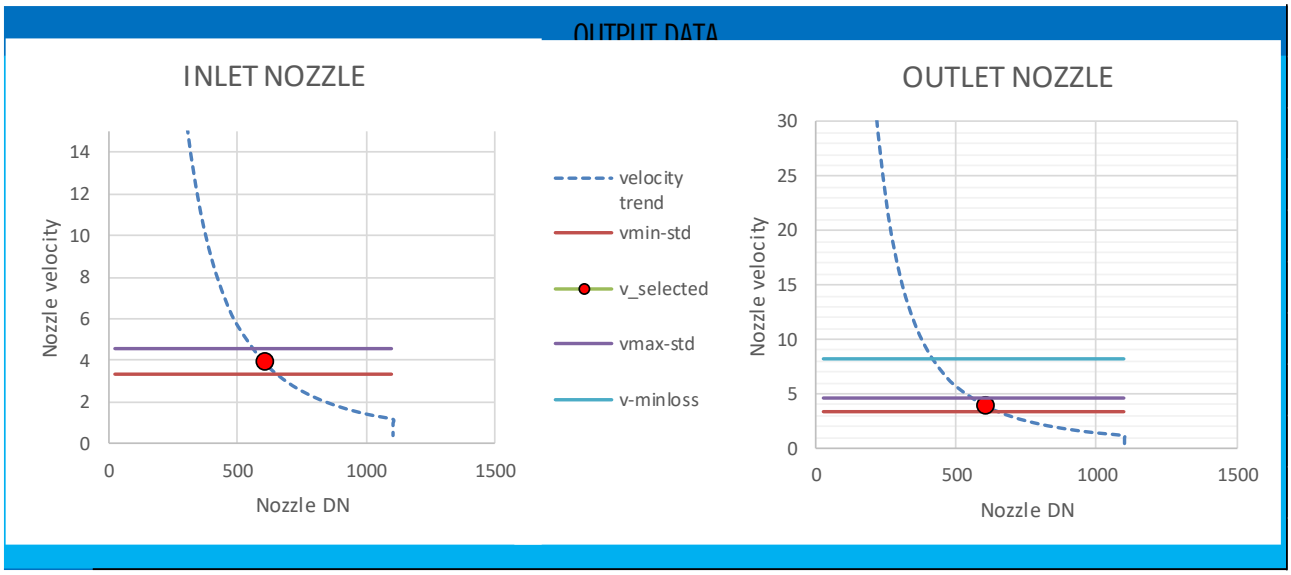

Figure 5. chart of velocity in the suction and discharge flange

6. Design recommendations.

The method described in the document is applicable for dimensioning the flanges of the pump and double checking any request from the Customer on the dimension of the connection between pump and piping of the plant.

The calculation losses criteria are affected by uncertainness related to the approximation of the diffusive duct so must be used for a high level method.

The dimension of the flanges strongly affects the overall dimension of the pump so must be carefully evaluated whenever a pump scaled from an existing hydraulic is selected. In this case the suggested flange dimension should be compared to the flange scaled from the source hydraulic: whenever the differences between the two solutions are remarkable a redesign of the hydraulic components is necessary and must be processed by Eng or R&D dpt.

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7. References

[1]- TMP and Gabbioneta handbook

[2]- Gulich "Centrifugal Pumps" ,4th.ed.

[3] Stepanoff "Centrifugal And Axial Flow Pumps" 2nd ed.

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